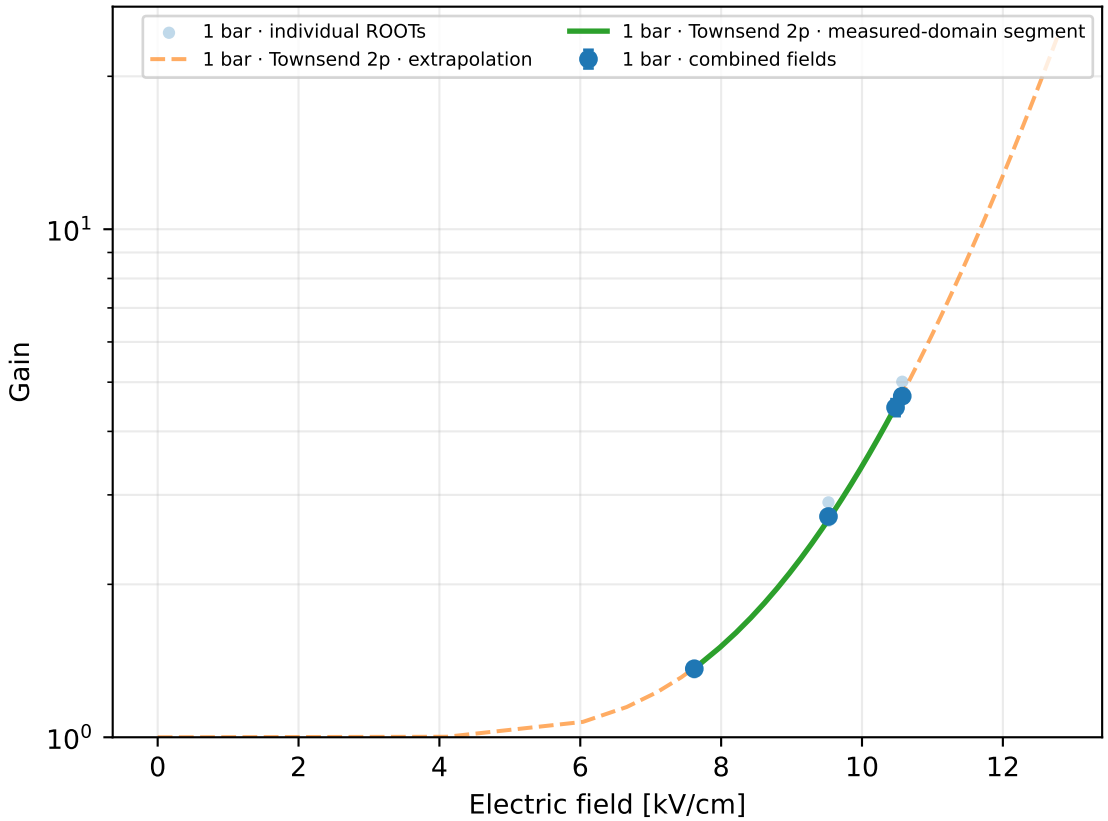
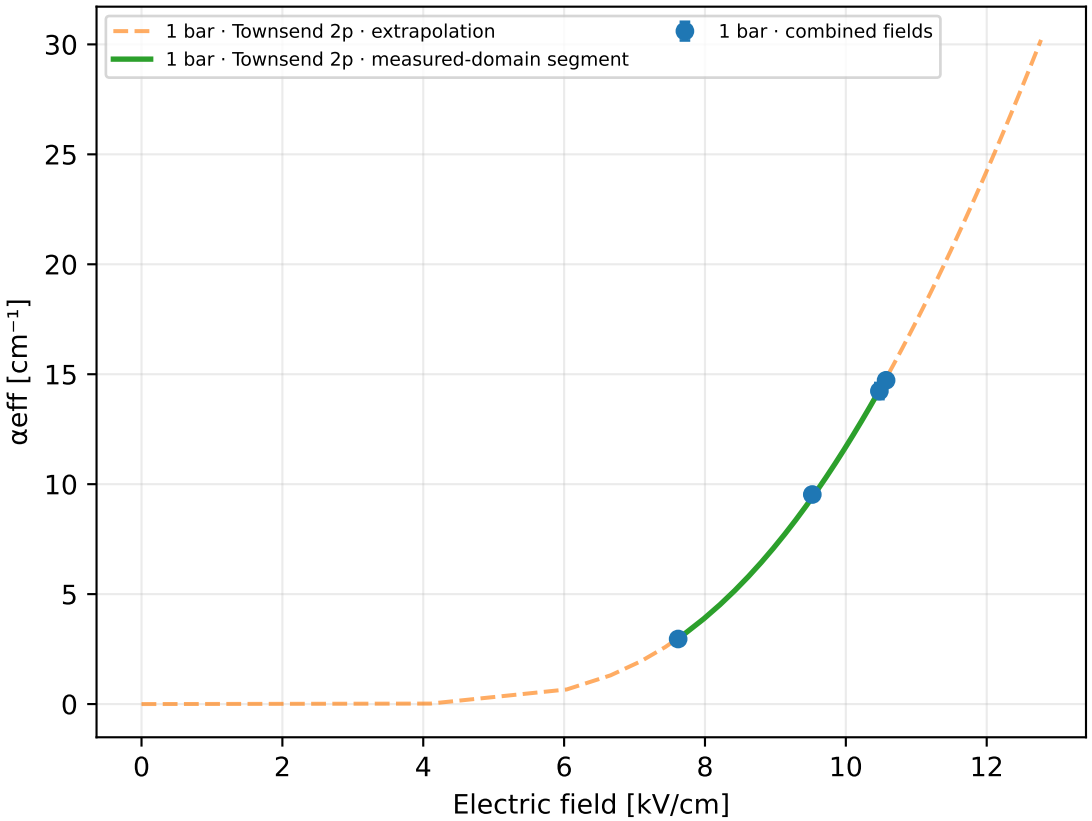
# ArCF4 · Ar 99.9 % / $CF_4$ 0.1 % · gap 1.05 mm

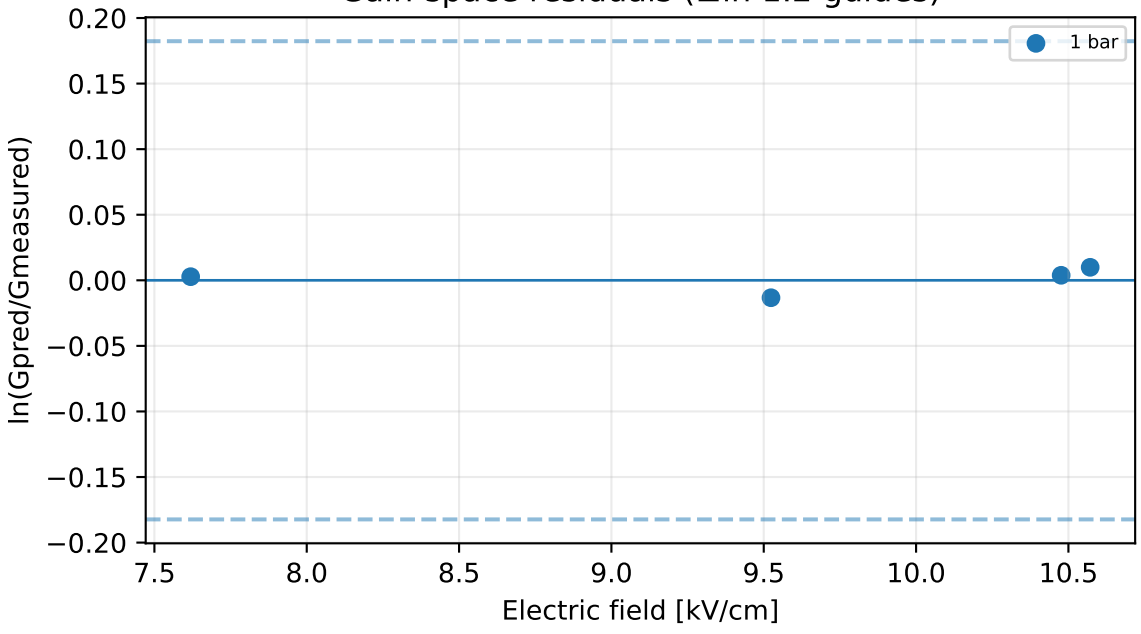
## Gain vs electric field



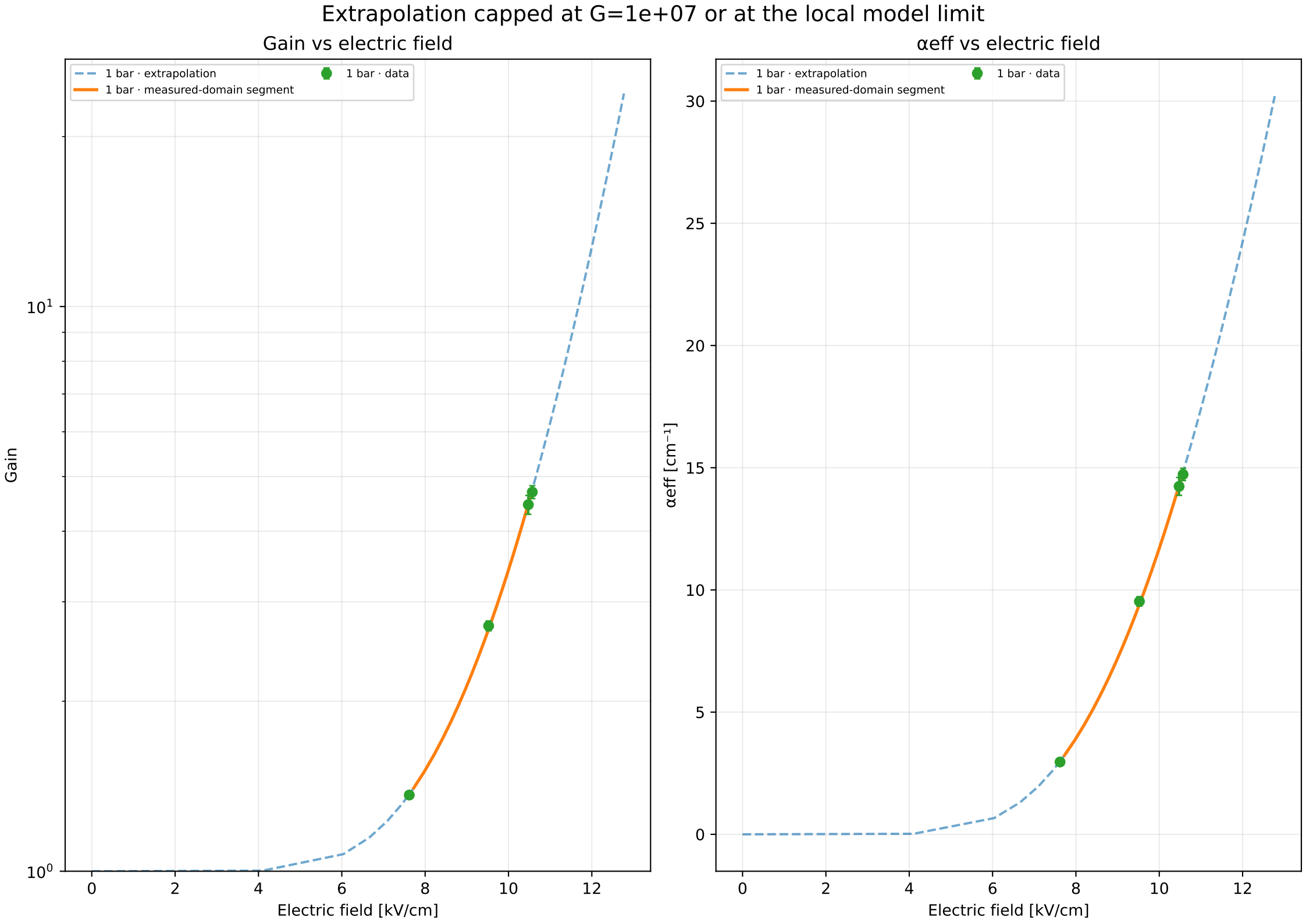

## Effective Townsend coefficient vs electric field




## Gain-space residuals (±ln 1.2 guides)




```
ROOT runs: 9
Unique physical points: 4
Pressures: 1 bar
Selected model: Townsend 2p
Status: rejected_poor_cross_validation
Training RMSE: ×1.009
Cross-validation median: unavailable
Cross-validation worst: unavailable
Reduced-field range: 7.619–10.57 kV cm⁻¹ bar⁻¹
A=925.61, B=43.709, m=0, n=1
Covariance condition: 1.7e+03
Warnings: poor_cross_validation, large_cross_validation_outlier
1 bar display limit: G=23.8 (local inversion limit)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
αeff vs electric field
1 bar · extrapolation
1 bar · measured-domain segment
1 bar · data
Gain
αeff [cm⁻¹]
Electric field [kV/cm]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

| p [bar] | E [kV/cm] | E/p | Gain | σGain | αeff [cm⁻¹] | npe | runs |
|---|---|---|---|---|---|---|---|
| 1 | 7.61905 | 7.61905 | 1.364375 | 0.0203 | 2.95901 | 1600 | 3 |
| 1 | 9.52381 | 9.52381 | 2.72 | 0.0547 | 9.52983 | 1600 | 3 |
| 1 | 10.4762 | 10.4762 | 4.458 | 0.171 | 14.2352 | 500 | 1 |
| 1 | 10.5714 | 10.5714 | 4.692727 | 0.123 | 14.7239 | 1100 | 2 |

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

| Candidate | k | AICc | CV median | CV worst | usable | warnings |
|---|---|---|---|---|---|---|
| Townsend 2p | 2 | -21.9 | — | — | no | poor cross validation, large cross validation outlier |